Credit Risk Classification: A Machine Learning Approach

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Abstract— *Credit risk assessment is a binary classification problem in which a lending institution must decide whether an applicant is likely to repay (good) or default on (bad) a loan. This paper presents a complete machine learning pipeline applied to the German Credit Risk dataset (1,000 applicants, 9 predictors). The pipeline covers exploratory data analysis, treatment of missing-not-at-random (MNAR) categorical variables, categorical encoding, feature scaling, a stratified 80/20 train–test split, and correction of class imbalance (70:30) using SMOTE. Three model families – Logistic Regression, a CART Decision Tree, and Random Forest – are trained with 5-fold cross-validation and compared using accuracy, precision, recall, F1-score, and ROC-AUC. The tuned Random Forest achieves the best test-set performance (Accuracy = 0.740, ROC-AUC = 0.720) and, according to permutation-based importance, identifies Checking_account, Duration, Saving_accounts, and Credit_amount as the most influential predictors of credit risk.*

Keywords— credit risk scoring, machine learning, random forest, logistic regression, decision tree, class imbalance, SMOTE, German credit dataset

I. INTRODUCTION

Credit risk assessment is a binary classification problem that aims to predict whether a loan applicant constitutes a *good* (repays the loan) or *bad* (defaults on the loan) credit risk, based on financial and demographic characteristics such as account status, loan term, loan amount, and age [1].

In the banking and financial sector, accurately quantifying this risk is of critical importance: loan defaults can generate substantial revenue losses and threaten the operational stability of lending institutions [1]. Traditional credit evaluation relied heavily on manual judgment and rule-based systems, which are susceptible to human bias and inconsistency. Machine learning methods offer data-driven, objective, and scalable alternatives that can process large volumes of applicant information to support more reliable lending decisions [2]. By analyzing customer attributes – such as age, employment status, savings account balance, credit amount, and loan duration – financial institutions can better understand the behavioral patterns associated with loan repayment and develop more effective risk-management policies [1, 3].

This paper implements and compares three classifiers – Logistic Regression, a CART Decision Tree, and Random Forest – for credit risk classification on the German Credit dataset, and evaluates them under the practical constraints of missing-not-at-random data and severe class imbalance.

II. DATASET

The dataset used in this project is the German Credit Risk dataset, a widely used benchmark in the credit-scoring literature [3, 9], publicly available on Kaggle [12]. It contains 1,000 samples and 9 feature variables, summarised in Table 1.

Table 1: Features used in the German Credit Risk dataset.

Feature	**Description**	**Type**
`Checking_account`	Status of existing checking account	Categorical
`Duration`	Duration of credit (months)	Numerical
`Purpose`	Purpose of the credit	Categorical
`Credit_amount`	Credit amount requested	Numerical
`Saving_accounts`	Savings account balance	Categorical
`Age`	Age of applicant (years)	Numerical
`Sex`	Gender of applicant	Categorical
`Housing`	Housing status	Categorical
`Job`	Job category	Categorical

The target variable, `Risk`, is imbalanced: 700 (70%) applicants are labelled *good* and 300 (30%) are labelled *bad*, as shown in Fig. 1.

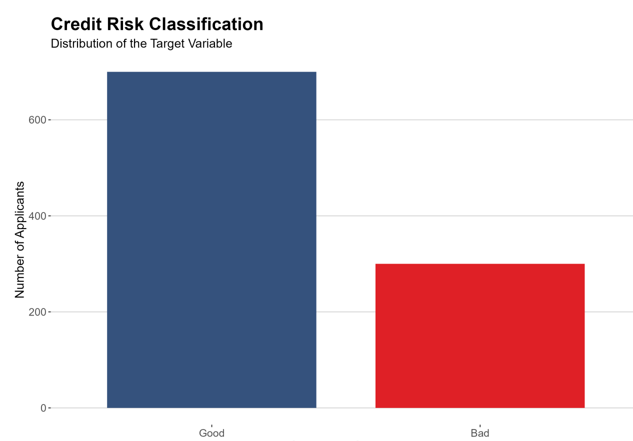

Figure 1: Distribution of the target variable (`Risk`): 700 *good*-credit vs. 300 *bad*-credit applicants.

A critical characteristic of this dataset is the high rate of missing values in two key financial variables:

Checking_account is missing for 39.4% of observations, while Saving_accounts is missing for 18.3%. In real-world banking contexts, the absence of account information may itself carry predictive signal, since applicants without formal accounts may represent a systematically distinct risk profile [10]. This missing-not-at-random (MNAR) characteristic required careful treatment during preprocessing rather than naive imputation.

III. RELATED WORK

Credit scoring has historically been dominated by traditional statistical methods such as logistic regression, linear discriminant analysis, and expert systems [4]. While interpretable and computationally efficient, these approaches assume linear relationships and variable independence – assumptions frequently violated in real-world credit data [1]. As a result, the research community has progressively shifted towards machine learning, which can capture nonlinear relationships and complex feature interactions more effectively [3].

Louzada et al. [3] conducted a systematic review of 187 credit-scoring papers published between 1992 and 2015, identifying Neural Networks (17.6%), hybrid methods (16.8%), ensemble methods (16.0%), SVMs (14.3%), and Logistic Regression (10.9%) as the most commonly applied techniques. Ensemble methods such as Random Forest and XGBoost have become the dominant category of newly proposed methods after 2012, accounting for over 21% of contributions [3, 8]. Across 30 studies using the German Credit dataset, reported accuracy values range from approximately 72.6% to 98.5%, underscoring the strong influence of preprocessing and experimental design [3]. Because misclassifying a bad applicant as good carries substantially greater financial cost, cost-sensitive evaluation frameworks are also explicitly recommended [5].

IV. METHODOLOGY

IV.1 Data Preprocessing

Missing values in Checking_account and Saving_accounts were recoded as an explicit "unknown" category rather than imputed numerically, preserving the potentially informative signal of missingness [10]. Ordinal encoding was applied to the two account-status variables; binary encoding was applied to Sex and the target Risk; one-hot encoding was applied to the nominal variables Housing and Purpose, expanding the feature space to 17 predictors. The three continuous features – Age, Credit_amount, and Duration – were standardized ($z = (x-\mu)/\sigma$). An 80/20 stratified train–test split preserved the original class ratio (train: 240 *bad* / 560 *good*; test: 60 *bad* / 140 *good*).

IV.2 Class Imbalance

The 70:30 class ratio biases naive classifiers towards the majority class, which is especially costly in credit scoring since a false negative (approving a truly bad applicant) is far more expensive than a false positive [5]. The Synthetic Minority Over-sampling Technique (SMOTE) [7] was applied exclusively to the training partition (over_ratio $= 0.8$, neighbors $= 5$), increasing the minority class in training from 240 to 448 observations while leaving the test set untouched.

IV.3 Models

Three complementary classification models were considered, representing linear, single-tree, and ensemble-tree learning paradigms. For each model, a baseline version using default hyperparameters was first fitted, followed by a tuned version obtained through systematic hyperparameter optimization.

- **Logistic Regression** models the log-odds of the positive class as a linear combination of features. Its tuned version uses Elastic Net regularization, with $\alpha = 1$ (Lasso) selected by the optimization procedure.
- **Decision Tree (CART)** recursively partitions the feature space to reduce Gini impurity. Its complexity parameter (cp) was optimized to control tree growth and avoid excessive pruning.
- **Random Forest** [11] combines multiple bootstrap-trained decision trees using random subsets of predictors at each split. The number of candidate variables (mtry) and number of trees (ntree) were considered during tuning.

IV.4 Hyperparameter Optimization

Hyperparameter optimization was performed using a grid-search strategy combined with 5-fold stratified cross-validation. A predefined grid of candidate hyperparameter values was evaluated systematically, and the configuration achieving the highest mean ROC-AUC across the validation folds was selected as the final tuned model. Stratification was used because the target variable is imbalanced (70% *good* and 30% *bad*), ensuring that each cross-validation fold retained approximately the same class distribution as the training data. This reduces the risk that model selection is driven by folds with disproportionately many observations from the majority class. ROC-AUC was selected as the optimization criterion because it evaluates discrimination across classification thresholds and is therefore more informative than accuracy alone under class imbalance.

For Logistic Regression, α was tested at 0, 0.5, and 1, while λ was tested using 20 values between 0.0001 and 1. For the Decision Tree, cp values between 0.001 and 0.049 were tested. For Random Forest, mtry was optimized, while ntree was set to 300. All models used the same stratified 5-fold cross-validation procedure, and the test set was kept separate from hyperparameter tuning.

V. RESULTS

All metrics were computed on the held-out 200-observation test set, which was never used during training or hyperparameter tuning.

Table 2: Model performance comparison (test set).

Model	Acc.	Prec.	Rec.	F1	AUC
LR – Base	0.690	0.787	0.764	0.775	0.684
LR – Tuned	0.685	0.781	0.764	0.773	0.685
DT – Base	0.700	0.700	1.000	0.824	0.500
DT – Tuned	0.695	0.797	0.757	0.777	0.687
RF – Base	0.725	0.785	0.836	0.810	0.720
RF – Tuned	**0.740**	**0.801**	**0.836**	**0.818**	**0.720**

The tuned Random Forest outperforms both the Logistic Regression and Decision Tree families across nearly every metric. The untuned Decision Tree is a degenerate case: with default `cp` it fails to split at all and always predicts the majority class, achieving 100% recall purely by predicting *good* for every applicant – reflected in an uninformative ROC-AUC of exactly 0.5. This illustrates why threshold-independent measures such as ROC-AUC are essential for model selection under class imbalance.

Fig. 2 shows the ROC curves for the three tuned models. The Random Forest curve dominates the Logistic Regression and Decision Tree curves across most of the false-positive-rate range, consistent with its higher AUC (0.720 vs. 0.685 and 0.687, respectively).

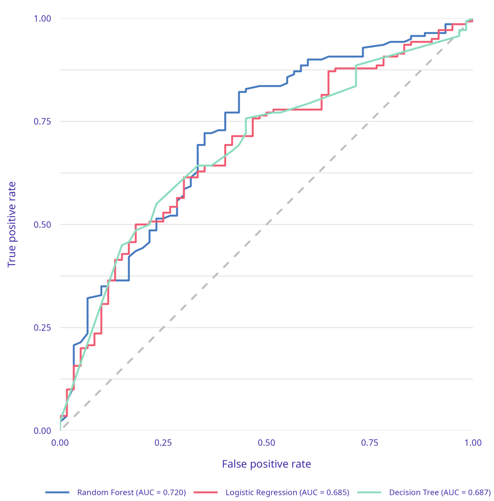

Figure 2: ROC curves for the three tuned models on the test set.

V.1 Feature Importance

Fig. 3 shows permutation-based feature importance for the tuned Random Forest, measured as one minus the AUC loss after permuting each feature. `Checking_account` is by far the most influential predictor, followed by `Duration`, `Saving_accounts`, and `Credit_amount` – consistent with financial intuition, since liquidity and loan-exposure variables directly signal repayment capacity. Demographic and one-hot-encoded `Purpose`/`Housing` variables contribute comparatively little individually.

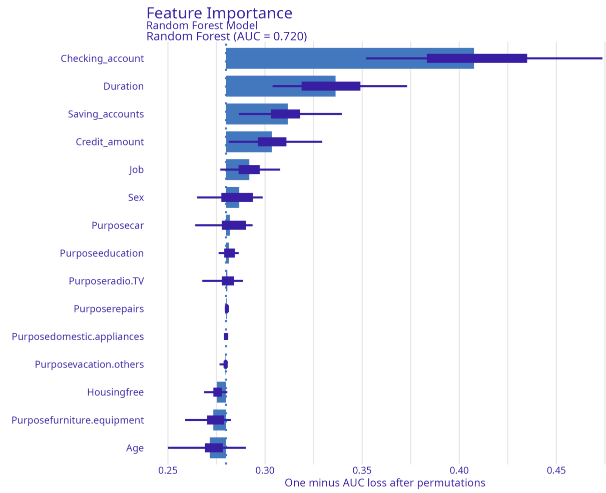

Figure 3: Permutation-based feature importance for the tuned Random Forest model.

VI. DISCUSSION AND CONCLUSION

This study implemented and compared three machine learning approaches for credit risk classification on the German Credit dataset, addressing MNAR missingness, mixed categorical/numerical features, and 70:30 class imbalance. Without correction and without a threshold-independent metric, a degenerate "always predict good" classifier can appear competitive on accuracy and even recall while carrying zero discriminative value, underscoring the importance of ROC-AUC as an evaluation criterion. The Random Forest ensemble consistently outperformed the single-model baselines, and account-status variables (`Checking_account`, `Saving_accounts`) – the two variables with the highest missingness – proved to be the most predictive features, validating the decision to encode missingness as an informative category.

The achieved ROC-AUC (0.720) trails the strongest results reported in the literature, most of which rely on gradient-boosted trees such as XGBoost or LightGBM with more extensive hyperparameter search [8]. Future work should extend the model set to gradient boosting with Bayesian optimization, incorporate cost-sensitive loss functions that penalise false negatives more heavily [5], and apply model-agnostic interpretability tools such as SHAP [6] to further validate feature-level decisions.

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